

WEEZE LAARBRUCH, Germany

WMO: 100001

Lat: 51.602N

Lon: 6.142E

Elev: 106

StdP: 14.64

Time Zone: 1.00 (EUC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.0	22.9	10.9	9.7	22.4	16.4	12.6	25.5	27.2	48.6	23.7	47.3	5.8	50	0.518

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.4	87.4	68.3	82.5	66.9	78.9	64.9	70.4	81.5	68.6	78.7	67.0	75.9	7.2	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.8	98.9	73.3	65.3	93.9	71.6	64.1	89.8	70.2	34.4	82.3	32.8	78.2	31.4	76.3	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
21.2	18.5	16.3	DB	14.4	95.9	8.2	3.4	8.5	98.4	3.7	100.4	-0.9	102.3	-6.9	104.8
			WB	13.5	74.4	8.0	1.8	7.8	75.7	3.1	76.7	-1.4	77.7	-7.2	79.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.7	37.2	38.7	44.3	50.9	57.3	63.0	66.3	65.5	59.9	52.6	44.3	39.6	(6)
	DBStd	12.26	8.05	7.14	6.81	6.80	7.04	6.01	6.15	5.49	5.15	6.43	6.91	7.79	(7)
	HDD50	1563	400	318	196	69	12	0	0	0	1	45	195	327	(8)
	HDD65	5104	863	736	640	423	257	109	56	59	168	386	621	786	(9)
	CDD50	2189	2	1	21	97	239	390	505	479	299	126	24	6	(10)
	CDD65	255	0	0	0	1	19	48	95	73	17	2	0	0	(11)
	CDH74	2250	0	0	0	29	188	434	843	619	127	10	0	0	(12)
	CDH80	691	0	0	0	2	41	123	296	199	30	0	0	0	(13)
(14) Wind	WSAvg	7.5	8.5	8.6	8.0	7.0	7.0	6.8	6.9	6.7	6.8	7.3	7.7	8.9	(14)
(15) Precipitation	PrecAvg	31.1	2.6	2.4	2.0	1.7	2.4	2.7	3.1	3.2	2.6	2.6	2.9	3.0	(15)
	PrecMax	40.5	5.6	5.1	4.4	3.5	4.5	7.9	5.6	6.9	6.4	7.8	4.8	6.3	(16)
	PrecMin	23.7	0.2	0.5	0.2	0.0	0.5	0.5	1.1	1.0	0.4	0.8	0.3	1.0	(17)
	PrecStd	3.9	1.3	1.2	1.2	0.8	1.0	1.5	1.3	1.7	1.6	1.2	1.2	1.2	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.6	58.9	68.0	77.4	85.7	89.7	95.1	91.3	84.3	75.0	62.9	57.1	(19)
		MCWB	52.0	48.1	54.4	58.4	66.3	70.8	70.2	69.3	66.5	63.0	58.2	52.8	(20)
	2%	DB	52.2	53.2	62.7	72.0	78.9	82.8	87.7	85.6	77.3	68.4	59.2	55.0	(21)
		MCWB	49.5	48.2	52.4	56.4	62.9	66.7	69.2	68.3	65.2	60.9	55.4	51.7	(22)
	5%	DB	50.0	51.0	58.6	67.8	74.9	78.8	82.1	80.6	73.1	64.5	56.9	52.1	(23)
		MCWB	47.9	47.2	51.3	55.3	60.9	65.2	66.9	66.4	63.5	59.5	53.7	49.5	(24)
	10%	DB	47.7	48.5	54.8	62.8	70.1	74.9	78.5	76.7	68.8	61.2	53.9	49.8	(25)
		MCWB	46.0	45.6	49.1	53.7	59.2	63.1	65.2	64.7	61.1	57.5	51.3	47.9	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.4	51.6	57.3	61.2	68.9	72.0	72.9	72.8	69.0	64.9	58.7	54.6	(27)
		MCDB	54.7	55.6	63.2	72.1	80.3	84.9	88.1	83.7	79.0	71.2	62.0	56.7	(28)
	2%	WB	50.4	49.5	54.2	58.3	65.7	69.1	70.3	70.0	66.7	62.1	56.0	52.1	(29)
		MCDB	51.9	51.9	60.1	68.5	75.3	78.2	82.0	79.8	74.3	66.4	58.6	54.1	(30)
	5%	WB	48.0	47.6	51.7	56.4	62.7	66.9	68.6	67.6	64.8	60.3	54.0	50.1	(31)
		MCDB	49.5	50.4	57.4	65.1	70.6	75.2	79.3	77.2	71.0	64.3	56.3	52.0	(32)
	10%	WB	45.7	45.8	49.8	54.6	60.2	64.8	66.7	65.9	62.8	58.5	51.8	47.9	(33)
		MCDB	47.2	48.4	53.9	61.9	68.5	72.7	75.7	74.0	67.7	61.6	54.2	49.3	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	8.7	10.0	14.2	17.3	17.4	17.2	17.4	17.5	15.5	12.9	9.5	7.8	(35)
		MCDBR	9.9	12.7	21.0	24.9	23.1	24.1	24.3	24.6	21.2	16.6	10.7	8.7	(36)
	5% WB	MCWBR	8.6	9.0	12.3	12.1	10.4	10.8	9.4	10.5	10.6	10.0	8.4	7.3	(37)
		MCWBR	9.1	11.4	17.5	21.0	19.9	20.0	21.5	20.7	17.8	15.0	10.2	8.3	(38)
(40) Clear-Sky Solar Irradiance	taub		0.331	0.353	0.384	0.418	0.422	0.420	0.430	0.421	0.394	0.367	0.344	0.326	(40)
		taud	2.369	2.337	2.258	2.198	2.226	2.248	2.225	2.273	2.338	2.397	2.375	2.344	(41)
	Ebn at Noon		213	241	254	259	264	265	260	256	249	233	205	192	(42)
		Edh at Noon	20	27	35	41	42	42	42	38	32	25	20	17	(43)
(44) All-Sky Solar Radiation	RadAvg		232	445	806	1299	1571	1689	1601	1361	995	585	274	177	(44)
	RadStd		32	85	118	143	136	141	198	122	107	51	32	22	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (84 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page